higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(Second Paper)
NQF LEVEL 4

25 FEBRUARY 2019

This marking guideline consists of 9 pages.

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SYMBOL	**EXPLANATION**
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
PL	Plot on graph
E	Explanation
S	Simplification
RT/RG/RD/RM	Reading from table/graph/drawing/document/map
F	Choosing correct formula
SF	Substituting in formula
O	Opinion
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off
R	Rounding
U	Unit
K	Knowing

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QUESTION 1

1.1	1.1.1	Party 1: 24 000 ÷ 200 = R120 per person✓ Party 2: 32 000 ÷ 200 = R160 per person✓	1A 1A		(2)
	1.1.2	They used a different caterer or a different menu.✓✓ (1 reason = 2 marks)	2R/J		(2)
	1.1.3	Variance is the difference between the actual and the projected amounts.✓✓	2K		(2)
	1.1.4	(a) Party 1: 7 550 – 10 000 = -2 450✓ Party 2: 16 320 – 15 500 = 820✓	1A 1A		(2)
		(b) They made a smaller profit than expected from Party 1.✓ They made a bigger profit than expected from Party 2.✓	1R 1R		(2)
	1.1.5	They should rather charge a higher price per person as it is more profitable.✓✓ (Any reasonable reason = 2 marks)	2R/J		(2)
1.2	1.2.1	Cost for 16 kℓ water:			
		6 kℓ = 6 × 4,56 = 27,36✓	1M		
		6 – 10,5 kℓ = 4,5 × 17,75 = 79,88✓	1M		
		10,5 – 16 kℓ = 5,5 × 24,93 = 137,12✓	1M		
		Total = 27,36 +79,88 + 137,12 = R244,36✓	1A		
		Cost for 400 kWh electricity:			
		400 × 1,92 = 768,00✓	1M		
		768,00 + VAT = R875,52✓	1M		
		Total cost = R244,36 + R875,52✓ = R1 119,88	1M		(7)
	1.2.2	It encourages people to use less.✓✓ (Any reasonable reason = 2 marks)	2R/J		(2)

1.3	1.3.1	Interest = R $8\,862,01 \times \frac{13,8}{100}$ = R1 222,96✓	1A	
		$= \frac{1\,222,96}{12}$ = R101,91✓	1A	
		Total = R8 862,01 + R101,91 = R8 963,92✓	1A 1M	
		Closing balance = R8 963,92 – R490✓ = R8 473,92		(4)
	1.3.2	Last payment = R190,83 + R2,19✓ = R193,02✓	1M 1A	(2)
	1.3.3	Payment = R(490 × 23) + R193,02✓ = R11 463,02✓	1M 1A	(2)
	1.3.4	It would be paid off sooner./The total amount paid would be less./The interest paid would be less.✓✓	2R/J	(2)
1.4	1.4.1	R61 910 + 31% of amount above R296 540✓	1RT	
		= R61 910 + $\frac{31}{100}$ × R(400 000 – 296 540)✓	1SF	
		= R61 910 + R32 072,60 = R93 982,60✓	1A	
			1M	
		Annual tax = R93 982,60 – R13 635✓ = R80 347,60✓	1A	
			1M	
		Monthly tax = R80 347,60 ÷ 12✓ = R6 695,63	1A	
		Yes, it is correct.✓		(7)
	1.4.2	The amount that he will have to pay will increase. ✓ He will move into the next tax bracket and pay at a higher tax rate.✓	2 R/J	(2)
				[40]

QUESTION 2

2.1	2.1.1	Mode✓ It shows them what the most popular choice is.✓	1A 1R/J	(2)
	2.1.2	6 groups of 200 people✓ **OR** 1 200 people	1A	(1)
	2.1.3	Pie charts✓ It will show the percentage/proportions of each choice.✓	1A 1R/J	(2)
	2.1.4	Chicken: $\frac{634}{6}$ = 106✓ Beef: $\frac{406}{6}$ = 68 On average more people chose chicken	1MA 1A 1MA 1A 1R/J	(5)
	2.1.5	The chance or likelihood of an event happening✓✓	2K	(2)
	2.1.6	$P(\text{chicken}) = \frac{104}{200} = \frac{13}{25}$	1 mark numerator 1 mark denominator	(2)
	2.1.7	$P(\text{beef}) = \frac{406}{1\,200} = 0{,}34$	1 mark numerator 1 mark denominator 1A	(3)
	2.1.8	$P(\text{chicken or beef}) = \frac{(360+66)}{600} = \frac{426}{600} = 71\,\%$	1 mark numerator 1 mark denominator 1S 1A	(4)
2.2	2.2.1	The sample must not favour one side in any way.	2K	(2)
	2.2.2	• The sample size must be big enough. • The sample must be representative of the whole population.✓✓ • An appropriate collection method must be used. • Collected data should be based on fact, not opinion.	4R/J	(4)

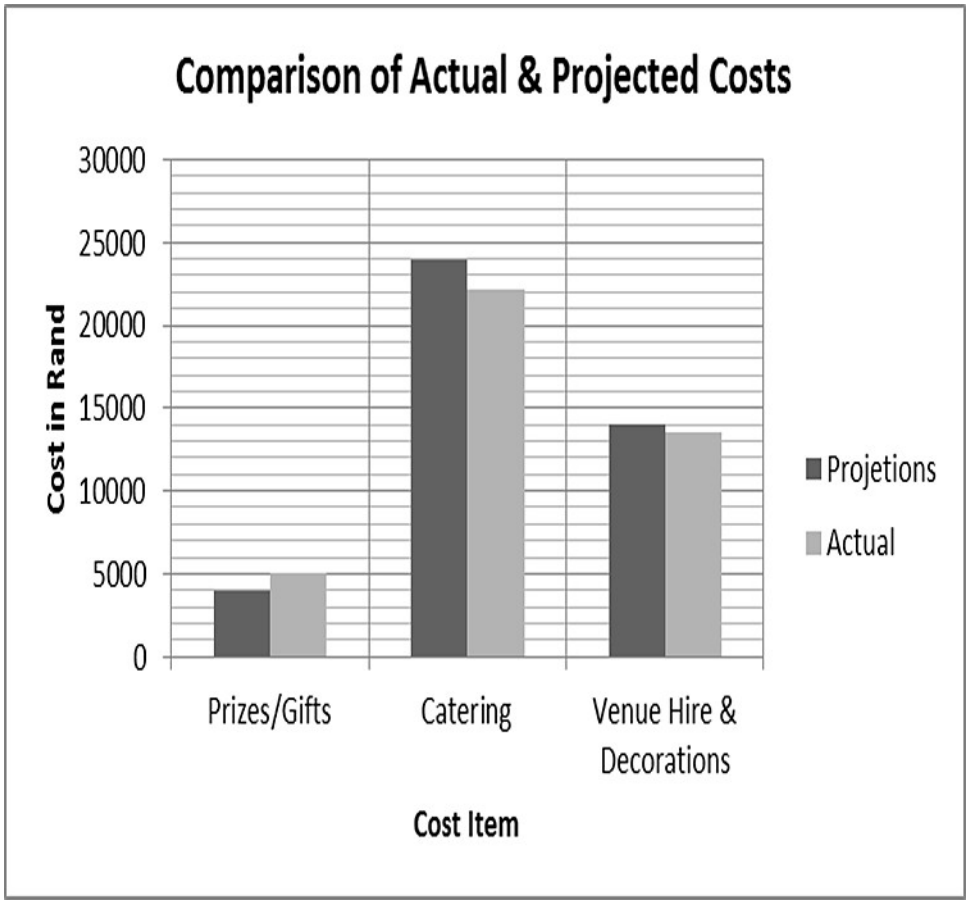
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(Any reasonable answer = 2 marks per answer)

2.3 2.3.1 Advantage: can see what is happening/true reflection of result✓ 1A
Disadvantage: time-consuming✓ 1A

2.3.2 Advantage: can reach a large number of people simultaneously✓ 1A
Disadvantage: sometimes difficult to get replies back✓ 1A
(Any 2 × 2) (4)

2.4



1 appropriate scale

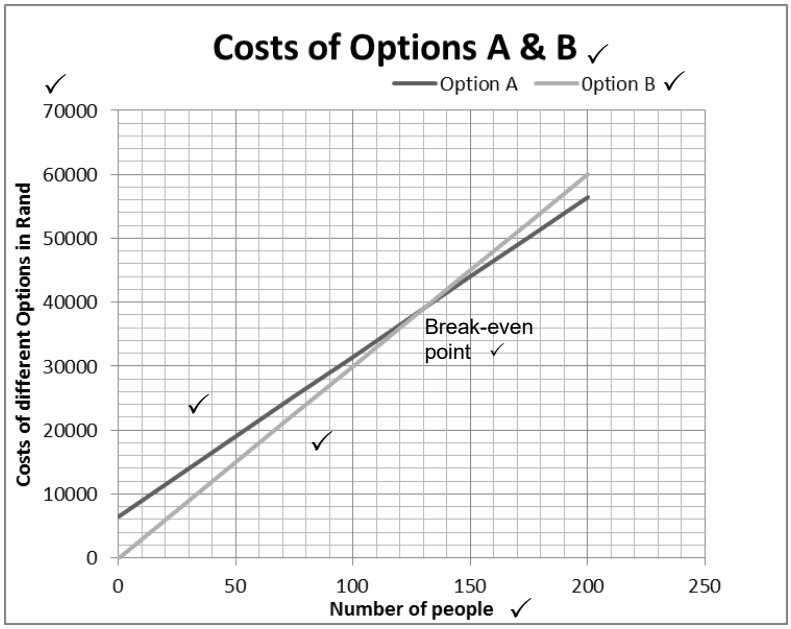
1 label both axes

1 suitable heading

3 correct bars

Appropriate scale✓
Labelling x- and y-axes✓
Suitable heading✓
Correct bars✓✓✓ (6)
[37]

QUESTION 3

3.1	3.1.1	A = 6 500 + (250 × 100)✓ = 31 500✓ B = 300 × 175✓ = 52 500✓ C = (56 500 – 6 500) ÷ 250 ✓ **OR** C = 60 000 ÷ 300✓ = 200✓ = 200✓	1M 1A 1M 1A 1M 1A	(6)
	3.1.2	Cost = 6 500✓ + (250✓ × number of people✓)	3A	(3)
	3.1.3	Cost = 300✓ × number✓ of people	2A	(2)
	3.1.4		1 appropriate scale 1 labels both axes 1 suitable heading 1 legend/ label line 1 break-even point 2 (1 each) correct line	(7)
	3.1.5	If more than 130 guests, choose option A.✓✓ If less than 130 guests, choose option B.✓✓	2R 2R	(4)
3.2	3.2.1	A = $\frac{12}{2}$✓ = 6 hours✓ B = $\frac{12}{1}$✓ = 12 people✓	1SF 1A 1SF 1A	(4)
	3.2.2	Inverse✓ – as the number of people increases,✓ the number of hours decreases✓	1A 2R/J	(3)
	3.2.3	• Time available✓✓ • Number of people available✓✓ • Budget for labour (Any 2 × 2)	2R/J 2R/J	(4)

[33]

QUESTION 4

4.1	4.1.1	999✓ km✓	1A; 1U	(2)
	4.1.2	N1✓ and N12✓	2 RM	(2)
	4.1.3	Distance = 102✓+ 80✓	2 RM	
		=182 km✓	1 A	(3)
	4.1.4	Total distance = 122 + 132 + 52 + 69 + 66✓✓	2 MA	
		= 441✓	1 A	(3)
4.2	4.2.1	Number of chairs = 23 × 8✓	1 M	
		= 184✓	1 A	
		There are more than 180 chairs.✓	1 E	(3)
	4.2.2	Size on plan = 3 m × 1 000 = 3 000 mm✓	1C	
		= 3 000 ÷ 200✓	1M	
		= 15 mm✓	1A	
		OR		
		Size on plan = 3 m ÷ 200✓	1M	
		= 0,015 m✓	1A	
		= 0,015 × 1 000 = 15 mm✓	1C	(3)
	4.2.3	80 ÷ 100 = 0,8 m✓	1C	
		Area of each overlay = 0,8 × 0,8✓	1SF	
		= 0,64 m^2✓	1A	
		Total area needed = 0,64 × 23✓	1M	
		= 14,72 m^2✓	1A	
		OR		
		Area of each overlay = 80 × 80 ✓	1SF	
		= 6 400 cm^2✓	1A	
		Total area needed = 6 400 × 23 ✓	1M	
		= 147 200 cm^2✓	1A	
		= 147 200 ÷ 10 000 = 14,72 m^2✓	1C	(5)
	4.2.4	Perimeter of each overlay = 80 cm × 4✓	1SF	
		= 320 cm✓	1A	
		Perimeter of overlay for 23 tables = 320 cm × 23		
		= 7 360 cm✓	1A	
		Total perimeter in metres = 7 360 ÷ 100✓	1 M	
		= 73,6 m✓	1C	
		Cost of satin ribbon = 73,6 × 3,20✓	1M	
		= R235,52✓	1A	(7)

4.3	4.3.1		Internal volume = 45 × 26 × 26,5✓	1SF	
			= 31 005 cm^3✓	1A	
			Capacity = 31 005 ÷ 1 000✓	1C	
			= 31,005✓ litres✓ (accept 31 litres)	1A	
				1U	(5)
	4.3.2	(a)	Number of 22 litre box = 80 ÷ 22✓	1M	
			= 3,64 ≈ 4 boxes✓	1R	
			Number of 30 litre box = 80 ÷ 30✓	1M	
			= 2,67 ≈ 3 boxes✓	1R	(4)
		(b)	Cost of transporting 22 litre box = 4 × 350		
			= R1 400✓	1 A	
			Cost of transporting 30 litre box = 3 × 420		
			= R 1 260✓	1 A	(2)
		(c)	The 30 litre box would be the most cost-effective option.✓	1R/J	(1)
					[40]
				TOTAL:	**150**

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